SOLUTION
SECTION-I

Single Correct Answer Type **6 Q. [3 M (–1)]**

1. Ans. (C)

Sol. Density of sphere is $\rho = \frac{m}{v} = \frac{3m}{4\pi R^3}$

$$\Rightarrow \frac{1}{\rho}\frac{d\rho}{dt} = -\frac{3}{R}\frac{dR}{dt}$$

Since $\Rightarrow \frac{1}{\rho}\frac{d\rho}{dt}$ is constant

$$\therefore \frac{dR}{dt} \propto R$$

Velocity of any point on the circumfrence V is equal to $\frac{dR}{dt}$ (rate of change of radius of outer layer).

2. Ans. (A)

Sol. $\frac{1}{a} = \frac{1}{4}v$

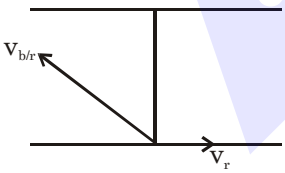
$$\frac{dt}{dv} = \frac{1}{4}v$$

$$\int_{t_1}^{t_2} dt = \int_{2}^{4} \frac{v}{4} dv$$

$$t_2 - t_1 = \frac{1}{4} \times \frac{1}{2}[16-4] = \frac{3}{2}$$

3. Ans. (C)

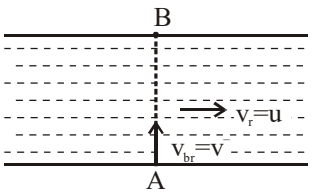
Sol.



$v_r > v_{b/r}$, so zero drift in not possible.
(C) boatman should be mantain velocity v of boat at certain angle greater than 90° with direction of river flow to minimize drifting & than walk rest of distance on other bank.

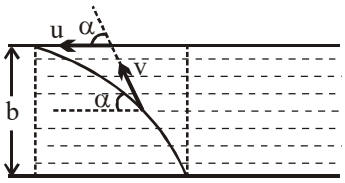
4. Ans. (A)

Sol.



In the river frame dock will move towards left with speed u

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$$ut_0 = v\int_0^{t_0} \cos\alpha\, dt \qquad \ldots (1)$$

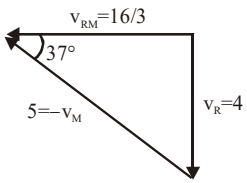
$$v - u\cos\alpha = -\frac{ds_{rel}}{dt}$$

$$\int_0^{t_0} (v - u\cos\alpha)\, dt = \int_b^0 -ds_{rel} \qquad \ldots (2)$$

$$vt_0 - \frac{u^2 t_0}{v} = b\,,\ t = \frac{bv}{v^2 - u^2}$$

5. Ans. (C)

Sol.



Multiple Correct Answer Type 4 Q. [4 M (–1)]

6. Ans. (A,C)

Sol. Using $C = \frac{1}{\sqrt{\mu_0 \in_0}}$ & $R = \sqrt{\frac{\mu_0}{\in_0}}$ we can check the correctness.

(A) $\mu_0 I^2 = \in_0 V^2$

$$\frac{\mu_0}{\in_0} = \frac{V^2}{I^2} = R^2$$

$\therefore R^2 = R^2$ correct

(B) $\in_0 I = \mu_0 V$

$$\frac{\in_0}{\mu_0} = \frac{V}{I}$$

$$\frac{1}{R^2} = R \text{ not correct}$$

(C) $I = \in_0 cV$

$$\frac{I}{V} = \in_0 c = \frac{\in_0}{\sqrt{\mu_0 \in_0}}$$

$$\frac{1}{R} = \sqrt{\frac{\in_0}{\mu_0}} = \frac{1}{R} \text{ correct}$$

(D) $\mu_0 CI = \in_0 V$

$$\frac{\mu_0 C}{\in_0} = \frac{V}{I}$$

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$$\frac{\mu_0}{\epsilon_0}\frac{1}{\sqrt{\mu_0\,\epsilon_0}} = R$$

$$\therefore \frac{1}{\epsilon_0}\sqrt{\frac{\mu_0}{\epsilon_0}} = R \Rightarrow \frac{R}{\epsilon_0} = R \text{ incorrect}$$

7. Ans. (A,B,C,D)

Sol. $\frac{dv(t)}{dt} = 6.0 - 3v(t) = 0$ (for terminal speed)

(A) $6 = 3v$

$v = 2$ m/s

(B) initial acceleration at $t = 0$ is $= 6 - 3(0)$

$= 6$ m/s^2

(C) $\frac{dv}{dt} = 6 - 3v \Rightarrow \int_0^v \frac{dv}{6-3v} = \int_0^t dt \Rightarrow \left|\frac{-1}{3}\ell n(6-3v)\right|_0^v = t$

$$\ell n\frac{6-3v}{6} = -3t \Rightarrow 6 - 3v = 6e^{-3t}$$

$6[1-e^{-3t}] = 3v \Rightarrow v = 2(1-e^{-3t})$

$v = 2(1 - e^{-3t})$ m/s

(D) $a = 3$ m/s^2

$3v = 6 - 3$

$3v = 3$

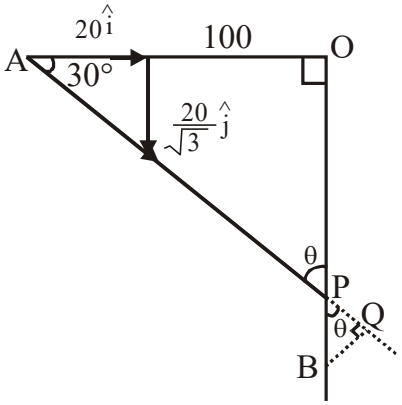
$v = 1$ m/s

8. Ans. (B, C)

Sol. In frame of B

$$\vec{v}_{A/B} = 20\hat{i} - \frac{20}{\sqrt{3}}\hat{j}$$

$$|\vec{v}_{A/B}| = \frac{40}{\sqrt{3}} \text{ m/s}$$



$$\frac{OP}{100} = \tan 30 = \frac{1}{\sqrt{3}}$$

$$OP = \frac{100}{\sqrt{3}}$$

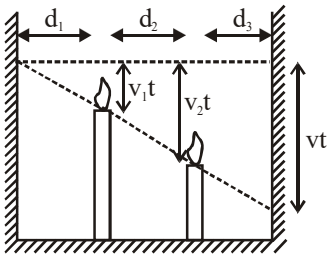
Minimum distance $\perp$ = BQ = BP sin θ

$$BP = \frac{200}{\sqrt{3}} \; ; \theta = 60°$$

In frame of B velocity of A is not along B, so they does not collide.

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9. Ans. (A,B,D)



Sol. $\frac{v_1 t}{v_2 t} = \frac{d_1}{d_1 + d_2}$

$\frac{v_2 t}{vt} = \frac{d_1 + d_2}{d_1 + d_2 + d_3}$

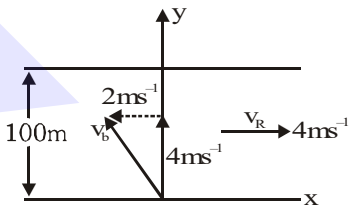
$v_2 = v\left(\frac{d_1 + d_2}{d_1 + d_2 + d_3}\right) = 1 \times \frac{30}{10} = \frac{1}{2} nm/s$

$v_1 = \frac{v_2 d_1}{d_1 + d_2} = v\left(\frac{d_1 + d_2}{d_1 + d_2 + d_3}\right)\left(\frac{d_1}{d_1 + d_2}\right) = 1 \times \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} mm/s$

$t_1 = \frac{200}{1/6} = 1200 \sec ; \; t_2 = \frac{200}{1/2} = 400 \sec$

10. Ans. (ABC)

Sol. Time of crossing $= \frac{100}{4} = 25s$



Absolute velocity of boat (i.e. w.r.t. ground)

$= |\vec{v}_{br} + \vec{v}_r| \; |\vec{v}_{br} + \vec{v}_r| = |-2\hat{i} + 4\hat{j} + 4\hat{i}| = (2\hat{i} + 4\hat{j}) = \sqrt{20} = 2\sqrt{5} ms^{-1}$

Drift of the boat = (4.2) × 25 = 50m

Linked Comprehension Type (2 Para × 3Q.) (1 Para × 2Q.)[3 M (-1)]
(Single Correct Answer Type)

11. Ans. (B)

12. Ans. (C)

13. Ans. (A)

Sol. (11 to 13)

$10 \text{ m/s}^2 = 5\frac{L}{T^2}$

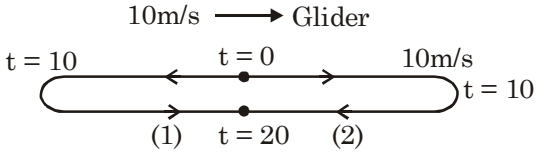
$\frac{1}{2} \times (500kg)(400m/s)^2 = 2000 \; ML^2T^{-2}$

(500 kg) (400 m/s) = 100 MLT^{-1}

Solve above equation to get L, T & M

14. Ans. (B)

Sol. Solve problem in the frame of glider. In glider frame speed of both air plane with be same.



in 10 minutes they with cover equal distance in glider frame & now they return to marker glider in next 10 minutes. (A) Option in correct & (B) is wrong (C) distance travelled for an observer in ground for airplane 1 & 2

$d_1 = (10 - 5) \times 600 + (10 + 5) \times 600$

$d_2 = (10 + 5) \times 600 + (10 - 5) \times 600$

d_1 & d_2 are equal

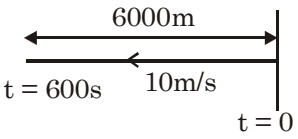
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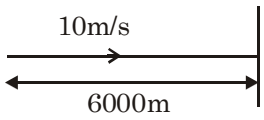
(d) for an observer on marker glider.

$d_1 = 10 \times 600 + 10 \times 600$, $d_2 = 10 \times 600 + 10 \times 600$

15. Ans. (A)

Sol. W.R.T. glider





$$t = \frac{6000}{10} = 600\,\text{sec} = 10\,\text{min}$$

16. Ans. (A)

Sol. No. of glider crossed for forward Journey

$$N_1 = \frac{6000}{500} = 12$$

No. of glider crossed for backward joruney

$$N_2 = \frac{6000}{500} = 12$$

$N = N_1 + N_2 = 24$

17. Ans. (C)

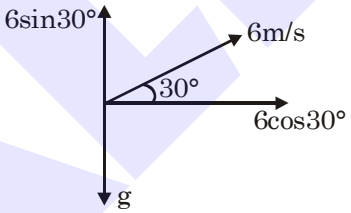
18. Ans. (D)

Sol. (17 & 18)

$$a = \frac{dv}{dt} = -2t$$

$v = -t^2 + 3$

$$s = 3t - \frac{t^3}{3}$$



when particle returns to ground

$s = 0,\ -\frac{t^3}{3} + 3t = 0,\ t = 3s$

So time of flight = 3s

Range = $6 \cos 30° \times 3 = 9\sqrt{3}$

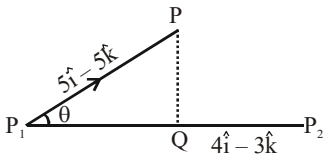
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SECTION-II

Numerical Answer Type Question **3 Q.[3(0)]**
(upto second decimal place)

19. Ans. 100.00

Sol.



$PP_1 = 5\sqrt{2}$

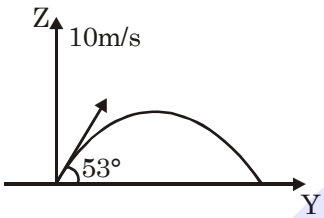
$\cos\theta = \frac{25+15}{\left(5\sqrt{2}\right)\times 5} = \frac{7}{5\sqrt{2}}$, $\sin\theta = \frac{1}{5\sqrt{2}}$

$PQ = (PP_1)\sin\theta = 5\sqrt{2}\times\frac{1}{5\sqrt{2}} = 1$

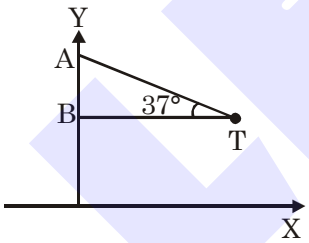
$t = \frac{PQ}{v} = \frac{1000}{10} = 100 \text{ sec}$

20. Ans. 105

Sol.



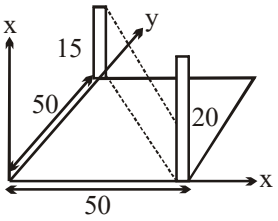
YZ plane is vertical plane



XY plane is horizontal plane

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 5}{10}} = 1 \text{ s}$

$v_x = \frac{x}{t} = \frac{50}{1} = 50 \text{ m/s}$

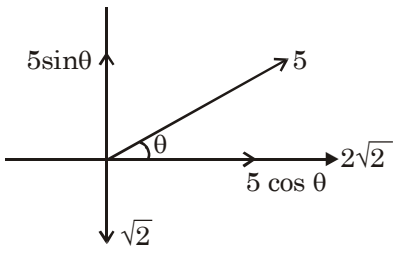


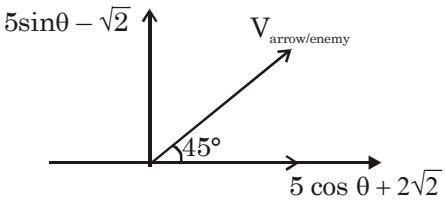
$v_y = \frac{y}{t} = \frac{50}{1} = 50 \text{ m/s}$

$v_z = \frac{z}{t} = \frac{5}{1} = 5 \text{ m/s}$

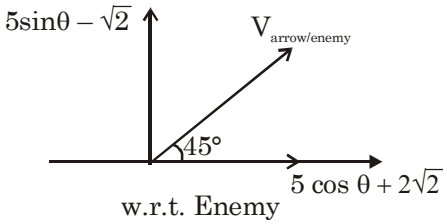
21. Ans. 82°

Sol.





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$5\cos\theta + 2\sqrt{2} = 5\sin\theta - \sqrt{2}$

$5\sin\theta - 5\cos\theta = 3\sqrt{2}$

$\frac{1}{\sqrt{2}}\sin\theta - \frac{1}{\sqrt{2}}\cos\theta = \frac{3}{5}$

$\sin(\theta - 45) = \sin 37°$

$\theta = 45° + 37°$

$\theta = 82°$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

22. Ans. 8

Sol. $t_A = \frac{100}{4} = 2.5\,\text{sec}$

$0 = v_B^2 + 2a(100)$

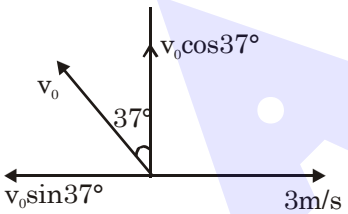
$0 = v_B + a(25)$ $\qquad a = -\frac{v_B}{25}$

$0 = v_B^2 + 2(-v_B)^4$

$v_B = 8\ \text{m/s}$ $\qquad a = -\frac{8}{25}\,\text{m/s}^2$

23. Ans. 5

Sol.



$v_{ball/g} = v_{ball/boat} + v_{boat/g}$

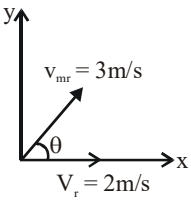
In horizontal direction velocity of ball w.r.t. ground = 0

$v_0 \sin 37° = 3$

$v_0 = 3 \times \frac{5}{3} = 5\ \text{m/s}$

24. Ans. 6

Sol.



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$x = (2 + 3\cos\theta) \times 2$

$y = (3\sin\theta) \times 2$

$\Rightarrow (x-4)^2 + y^2 = (6)^2$

25. Ans. 5

Sol. For collision $u = \sqrt{u_B^2 + u_C^2} = \sqrt{\left(\frac{72}{5} \times \frac{5}{18}\right)^2 + \left(\frac{54}{5} \times \frac{5}{18}\right)^2} = \sqrt{4^2 + 3^2} = 5$ m/s

SECTION-IV

Matrix Match Type (4 × 5) — 2 Q. [8 M (for each entry +2(0)]

26. Ans. (A) P,S,T; (B) P,Q,R,T; (C) P,S,T; (D) P,Q,R,T

27. Ans. (A) R; (B) R; (C) Q,S; (D) P,S

Sol. **For (A)**, $v^2 \propto x \Rightarrow$ Acceleration = constant, **For (B)**, $v \propto t \Rightarrow \frac{dv}{dt}$ = constant

For (C), $v \propto t^2 \Rightarrow \frac{dv}{dt} \propto t$, **For (D)**, $v \propto x \Rightarrow \frac{dv}{dt} = v\frac{dv}{dx} \Rightarrow \frac{dv}{dt} \propto v$

Subjective Type — 3 Q. [4 M (0)]

28. Ans. 21.75 metre.

Sol. Distance covered by the car during the application of brakes by driver –

$s_1 = ut = \left(54 \times \frac{5}{18}\right)(0.2) = 15 \times 0.2 = 3.0$ meter

After applying the brakes; $v = 0$ $u = 15$ m/s, $a = 6$ m/s^2 $s_2 = ?$

Using $v^2 = u^2 - 2as \Rightarrow 0 = (15)^2 - 2 \times 6 \times s_2 \Rightarrow 12\, s_2 = 225 \Rightarrow s_2 = \frac{225}{12} = 18.75$ metre

Distance travelled by the car after driver sees the need for it $s = s_1 + s_2 = 3 + 18.75 = 21.75$ metre.

29. Ans. $\tan^{-1}(1/3)$

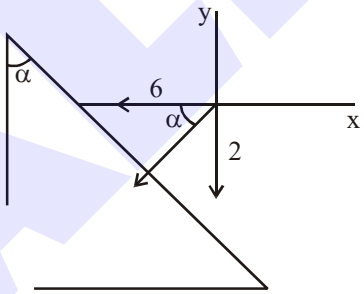
Sol. $\vec{v}_{a/g} = 6\hat{i}$

$\vec{v}_{r/a} = -2\hat{j}$

$v_{r/c} = -6\hat{i} - 2\hat{j}$

$\tan\alpha = \frac{2}{6}$

$\alpha = \tan^{-1}\left(\frac{1}{3}\right)$



30. Ans. (i) $\frac{u^2 \sin 2\alpha}{g\cos\theta}$ (ii) $v = \frac{u\cos(\alpha+\theta)}{\cos\theta}$

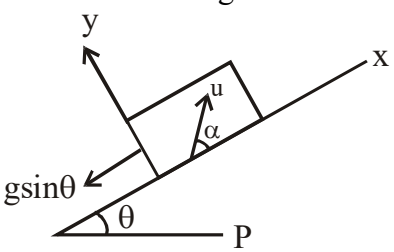
Sol. (i) box is sliding without friction, so acceleration of block will be g sin θ

$\vec{u}_{p/b} = u\cos\alpha\hat{i} + u\sin\alpha\hat{j}$

$\vec{a}_{P/G} = -g\sin\theta\hat{i} - g\cos\theta\hat{j}$

$\vec{a}_{B/G} = -g\sin\theta\hat{i}$

$\vec{a}_{P/B} = \vec{a}_{P/G} - \vec{a}_{b/G} = -g\cos\theta\hat{j}$



So there no acceleration in x-direction in box frame , so velocity will be constant in x-direction.

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For time to hetting ground
On y-axis

$$S = ut + \frac{1}{2}at^2$$

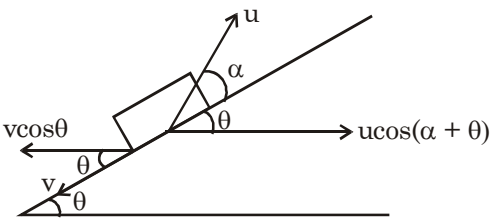
$$O = u\sin\alpha t - \frac{1}{2}g\cos\theta t^2$$

$$t = \frac{2u\sin\alpha}{g\cos\theta}$$

Distance between P & Q on x-axis

$$S = ut = u\cos\alpha\left(\frac{2u\sin\alpha}{g\cos\theta}\right) = \frac{u^2\sin 2\alpha}{g\cos\theta}$$

(ii)



In horizontal direction there is no acceleration. So velocity in horizontal direction will remain constant and as per given in question there is no displacement in x-direction so velocity in x-direction will be zero. $u\cos(\alpha+\theta) - v\cos\theta = 0$

$$v = \frac{u\cos(\alpha+\theta)}{\cos\theta}$$

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